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Is the cat dead, dead dead, alive, truly alive or playing jazz?

or, using the quantum framework to disambiguate language

Contents

1 Introduction

The goal of this paper is to present if and how notions and strategies proper to quantum mechanics can be applied to linguistic analysis, in an attempt to quantify and model lexical ambiguities. We mainly focus on the works of Wang, Sadrzadeh, Abramsky and Cervantes [8], which notably revolve around quantum-like entanglement between multiple words and the contextuality of their meaning.

2 Linguistic ambiguity

First of all, let us clarify the matter at hand. There exist two main types of linguistic ambiguities. Syntactical and lexical. The first solely relies on unclear arrangement of words, otherwise clear themselves. A classic example being "I see a man using my telescope". Such a sentence can be interpreted in two ways, depending on who the "using my telescope" adjoint is attached to. Either the man has it, or I do and using it allows me to see the man. This type of ambiguity can be straightforwardly formalised, using general syntactical tree structure. The difference of meaning can then be seen with ease as both interpretations result in completely different graphs. This method is crucially the result of the compositional approach to meaning, which claims meaning of sentences or words can be derived from the smaller fragments they are made of (in the case of sentences, those fragments being words, and in the case of words, those fragments being morphemes). One key aspect to identify here, is that this framework fundamentally struggles with giving meaning to those more basic elements. If a linguistic atom cannot be split further, how can we derive its meaning without simply imposing it? To avoid this call to fundamentalism, lexical meaning is, contrary to that of sentences, generally understood through the distributional framework. Instead of deriving meaning through composition, the distributional approach stems from the idea that "words that often occur in the same context have similar meanings and that one can know a word by the company that it keeps"[7]. As such, the meaning of a word, although less tangible, turns to a web of interconnections to other ones that seam to occur simultaneously in certain contexts. One easy way to visualise this, is to think of meaning as the ensemble of words you would use to describe something. A tiger is a "big cat, orange, dangerous, found in jungles". The distributional framework, although this is a simplification, would then assign the web of all words, with stronger links to the aforementioned commonly occurring words as the meaning of "tiger". This makes distributional analysis particularly unfit to understand sentences, as entire sentences rarely appear multiple times, and even more rarely to a statistical significant degree. However, there is one crucial thing to notice here. Neither of these frameworks seem to be able to analyse lexical ambiguities.

Indeed, lexical ambiguity is more subtle than its syntactical counterpart, and traditionally harder to explicitate formally[1]. Furthermore, it is of multiple nature, since "[at] the lexical level, one can distinguish between two broad types of ambiguity: homonymy refers to cases in which, due to some historical accident, words that share exactly the same spelling and pronunciation are used to describe completely distinct concept [...] On the other hand, the senses of a polysemous word are usually closely related with only small deviations between them"[5]. For an example of homonymy, look no further than the title of this essay, as the word "cat" can both refer to the furry domestic animal as well as, in jazz slang, to a man. Both meanings, a term crucially different from "sense", though we will gladly bundle both under the umbrella term "interpretation", followed separate etymological paths that simply ended up converging in spelling. On the other hand, both "dead" and "alive" are great examples of polysemy. Both can be used in a literal sense, but find another use in meaning "tired" or "thriving" respectively[2]. Those alternative senses are more intertwined with the original one, as they stem from it, the first from an exaggerated use of "dead", and the second a more enthusiastic use of "alive". Now, notice how neither compositionality, given its struggles with understanding unsplittable words, nor distributionality, which doesn't separate occurrences of words by their meaning, as meaning is precisely what we wish to gather, can be used to explain lexical ambiguities. This is where quantum theory might help.

[1] Although some have used the distributional framework for disambiguation, see [6].

[2] It could be argued that this divergence is instead due to pragmatics, and to the fact that someone saying "I'm dead" cannot realistically be so literally. This distinction is however not important, though it highlights the complexity of separating context-induced meanings from "formal" ones. A tried and tested non-pragmatic alternative example of a polysemous word could be the word bank, meaning both the building and the financial establishment.

3 Quantum strategy

What if ambiguous words were akin to superimposed quantum states? This is the main idea behind Wang et al.[8], in continuation to [5], and, indeed, if one could view standard unambiguous words as pure states, one might be able to create some understanding of lexical ambiguity from there. Namely, homonyms would be perceived as a superimposition of said pure states[3], one that could easily collapse under measurement. But what measurement? Well, reading, or hearing, or even more generally: interacting with the word. Engaging with an ambiguous word causes a collapse of its meaning. Indeed, it does not seem possible for an agent to hold all the possible meanings of a word true at once. Instead, we select one, based on the context. If I say I saw a cat on the street, you would more likely than not imagine the animal. If I say I saw a cat in a jazz club, that reading might not be your preferred one[4]. This apparent "contextuality" of language the authors resume in the following way: "given that a certain interpretation of a word is selected within a certain context, can we use this information to deduce how the same word may be interpreted in a different context (e.g. in different phrases) in the corpus? Our intuition is that this is not the case"[8]. This context dependency is the reason why one of the goals of the method we are about to present is to show its quantum contextuality. However, note that both notions are fundamentally different, proper to two different fields, the later being in some sense stronger, though we will come back to that.

This method crucially sits at the frontier between compositionality and distributionality. Indeed, we are still looking for statistical co-occurences, similarly to the distributional work in [6], but inside a certain structure, a certain pairing. This allows us to avoid the previously encountered "[...] issue with this framework [...] that the grammatical structure of phrases and sentences is ignored and the focus is mainly on large-scale statistics of data."[8]. Given the known tendency in linguistics to relegate issues in one's approach to a different subfield (think of the numerous debates around focus, which syntax claims is for semantics to solve and semantics for syntax to handle), the perspective of constructing a method reliant on two traditionally divergent approaches is a worthy goal in itself. But, to see if it actually works, let us first describe the process.

4 Possibilistic contextuality

Take the typical quantum experiment where Alice and Bob are each given entangled particles in superposition. Wang et al.[8] propose an analogous situation in language, where entanglement is a syntactic link, superposition is homonymy, and particles are words. For a concrete example, we consider one of their cases, take the words *coach, boxer, lap* and *file*. Each one of those words is ambiguous. We index their meanings the way described in

[3]Note that polysemous words can also be explained in a different way, see [5], more consistent with the general linguistic theory. This reading claims polysemous words to be "bundles" of vector states, which aligns more clearly with the theory (given homonymy is the inability of differentiating words, whereas polysemy is inner confusion bound to one and only lexical unity), but, as it strays away from the method here presented, we will ignore said differences and treat polysemy as superimposition too. To avoid confusion, all following examples in this paper are homonyms.

[4]We will later briefly look at seemingly unambiguous ambiguous words. Most people would only think of the furry animal in this example, as the other reading is jargon, or in-group slang the agent might not be aware of. The point, however, still holds.

word	meaning 0	meaning 1
coach	sports trainer	bus
boxer	athlete	dog breed
lap	run	drink
file	document	smoothen

Table 1: Sample of ambiguities

noun	verb	(0,0)	(0,1)	(1,0)	(1,1)
coach	lap	**1**	**1**	**1**	**0**
coach	file	**1**	**1**	**0**	**0**
boxer	lap	**1**	**1**	**1**	**1**
boxer	file	**1**	**1**	**0**	**0**

Table 2: Possibilistic combinations

Table 1. Now, combine all these words in a subject-verb structure (this is where compositionality comes into play) and for each of their respective meanings, ask yourself "does this reading make sense?". Does it make sense, for instance, for a dog to drink? does it make sense for a bus to document something? If yes, we note the combination with a **1**, and if not with a **0**.[5]. This, according to the authors[6], gives us the results of Table 2. As mentioned previously, one of our goals with this framework is to achieve contextuality, but that title alone is not specific enough, as we differentiate between three kinds: probabilistic, possibilistic and strongly possibilistic. This nomenclature comes from Abramsky's work, see [2], who summarised the criteria each type must obtain[7]. For now, we are only concerned by possibilistic contextuality, which Table 2 happily satisfies. Results are possibilistically contextual if "[there] exists a local assignment [...] which cannot be extended to a compatible family of local assignments"[8][2]. A local assignement is the function determining with which meaning we should interpret a certain word. In our case, a family of local assignments is said to be compatible if we assign a set meaning to each word such that the syntactical combinations of those meanings are meaningful. For example, $coach \to 1$ is the assignment setting the meaning of the word "coach" to "bus", and $coach \to 0$ the one setting it to "sports trainer". The set $\{coach \to 0, boxer \to 0, lap \to 0, file \to 0\}$ is a compatible family of local assignments. To verify it, we simply check that the cases corresponding to $(coach \to 0, lap \to 0)$, $(coach \to 0, file \to 0)$, $(boxer \to 0, lap \to 0)$ and $(boxer \to 0, file \to 0)$ all hold a **1**. On the contrary, what makes this array contextual is the impossibility of creating such a family containing certain attested combinations. One can check that the combination $(coach \to 0, lap \to 1)$, despite being considered possible, cannot be contained in any family without said family to be inconsistent. This is precisely possibilistic contextuality, which in our linguistic case can be reduced to the impossibility of assuming a certain set meaning for every ambiguous word without running into any impossible phrasings. Then,

[5]To avoid confusion with the indexes of meanings, we embolden those numbers.

[6]We will see later that this method induces confusion, as some readings deemed possible were never found in corpus, and some deemed impossible were instantiated, albeit via metaphors and figures of speech.

[7]This is the origin of the problem in the previous footnote. Those types of contextuality are of gradual strength, where strongly possibilistic entails possibilistic, which itself entails probabilistic. Misalignment between the possibilistic and statistical tables risks breaking this chain (and it does).

[8]Though we will not use the term here, note that strong possibilistic contextuality simply requires that no local assignement can be extended.

noun	verb	(0,0)	(0,1)	(1,0)	(1,1)
coach	lap	2/11	7/11	2/11	**0**
coach	file	43/44	1/44	**0**	**0**
boxer	lap	11/53	22/53	8/53	12/53
boxer	file	35/54	19/54	**0**	**0**

Table 3: Probabilistic combinations

should we simply stop here? We have indeed found a case of contextuality, implying there is at least one situation where the framework can be applied, right? Well, to ensure that this system actually acts similarly to quantum behaviour, we might need to take a small detour to another key concept: non-signallity. Take the original instance of Alice and Bob's experiment. We would call the setup non-signalling if one key criterion obtains: no matter what Alice observes, this should not impact what measurements Bob can conduct, and inversely. And, as it turns our, one of the consequences of Bell's Theorem is that any quantum system needs to be non-signalling to ensure proper behaviour, see [2]. The implications this has on our concerns may be disputable: indeed, no equivalent theorem has been proven of language and, furthermore, "certainly there is no reason to assume it"[8]. This is, nevertheless, a requirement of the quantum framework and, as such, using said framework on language requires the considered systems to be non-signalling[9,10]. Notably, our case in Table 2 does not satisfy that requirement. Suppose the meaning of "coach" collapses to that of "bus" (this is the equivalent of Alice getting their measurement result), then, it is simply impossible for "file" to follow, as none of its meanings would be compatible (see that once we fix *coach* $\rightarrow$ 1, both file cases are **0**). This shows the system is signalling, and thus does not behave in quantum-like ways[11]. We must then find some other approach.

5 Probabilistic contextuality

As a response to this issue, Wang et al.[8] changed focus and turned their heads towards probabilistic contextuality, hoping to find an instance both contextual and non-signalling. This strategy diverges in two main aspects from the possibilistic outlook: first, we replace our question "Does this make sense?" with data analysis of linguistic corpus. That is, we look for attested occurrences of the word pairs we consider, instead of native gut feeling. Secondly, we replace our **1** and **0**'s with statistical probabilities[12]. The result can be seen in Table 3, keeping the same words as in our previous example for simplicity. This probabilistic approach gives us more information to work with. Indeed, instead of simply having the possibility of a certain reading, we now know the ratio between interpretations of a single pairing[13]. A consquence of these alterations is that, now, contextuality of the system has to

[9] Even if the entire language itself is not

[10] This requirement can be softened through the CbD framework we consider later, but should still apply to standard quantum theory applications.

[11] Strong possibilistic contextuality could be used here as a way to enforce applicability of the framework, as it naturally entails non-signallity, see [2]. Strengthening the contextuality requirement would however reduce the amount of cases the framework is applicable to, which explains why it is not a solution considered by the authors.

[12] Please note that a consequence of this is that all probabilities on a same line add up to 1.

[13] Note that, once again for readability, unattested interpretations will be marked with a bold **0**.

be checked slightly differently. Indeed, we are not working with possibilistic but probabilistic contextuality, a version significantly weaker. This means that, though some cases did not behave contextually previously, they still might with this statistical approach. To check probabilistic contextuality, we need to work with Bell inequalities, see [2]. The general idea is still similar to what we previously presented, in that we want to prove non-classical behaviour, i.e. statistical distribution that cannot be explained via the work of an external source (of a global assignment, if you will). Bell proved that a certain inequality, when violated, showed quantum-like contextuality of the system. Here is the method to calculate it: take a few (one for each line of the table) logical "unjointly satisfiable"[2] propositions, then add up the probabilities of the cases (bound to their respected lines) that satisfy said propositions. This sum must be equal or below the number of lines in table, minus 1, for the system to be classically explainable, given any particular set of formulas. Contextuality is ensured when the inequality fails. Here, given our example, are the formulas one might consider:

$$\begin{aligned}
\varphi_1 &= (\mathit{coach} \to 1 \wedge \mathit{lap} \to 1) \vee (\mathit{coach} \to 0 \wedge \mathit{lap} \to 0)\\
\varphi_2 &= (\mathit{coach} \to 1 \wedge \mathit{file} \to 1) \vee (\mathit{coach} \to 0 \wedge \mathit{file} \to 0)\\
\varphi_3 &= (\mathit{boxer} \to 1 \wedge \mathit{lap} \to 1) \vee (\mathit{boxer} \to 0 \wedge \mathit{lap} \to 0)\\
\varphi_4 &= (\mathit{boxer} \to 0 \wedge \mathit{file} \to 1) \vee (\mathit{boxer} \to 1 \wedge \mathit{file} \to 0)
\end{aligned}$$

The inconsistency of those formulas is easy to check. We then assign to every φ_i the probability p_i for it to obtain, in our case:

$$\begin{aligned}
p_1 &= 2/11 + 7/11 = 9/11\\
p_2 &= 43/44 + 0 = 43/44\\
p_3 &= 11/53 + 12/53 = 23/53\\
p_4 &= 19/54 + 0 = 19/54
\end{aligned}$$

Our Bell inequality to transgress is then:

$$\sum_{i=1}^{4} p_i \leq 3$$

After adding all p_i's together, we get $\sim 2,58$, which is lower than our assumed threshold of 3 (the number of lines, minus one). This shows that this particular choice of formulas does not prove contextuality of the system. However, in our case, that is not needed, as we have already proven possibilistic contextuality of the support[14] of that system, and as such, we know the model is contextual.

This approach nonetheless does not solve the none signallity issue we faced. Indeed, Wang et al.[8] observe that it even makes it harder to attain. If we previously only had to check for possibility of measurement, we now must also verify that statistical distributions are shared between interpretations. And, indeed, Wang et al.[8] observe that many of the considered systems still refuse to abide to non-signallity. Considering our example, let us suppose we fix $\mathit{coach} \to 0$, then the sum of probabilities between a measurement with "lap" ($2/11+7/11 = 9/11$) is not the same as with a measurement

[14]The support is simply the same table as our probabilistic results, with **1** inserted wherever a non-zero statistic has been observed. In short, it is the underlying possibilistic structure of a probabilistic model. This explains why probabilistic contextuality naturally follows from its possibilistic counterpart

through "file" $(43/44 + 1/44 = 1)$. But an identical distribution is what defines possibilistc non-signallity, showing it still does not hold in our case. And as such, one major issue arises: the instability of the corpus. It is extremely unlikely for probabilities to magically show patterns in their repeated sums. Furthermore, suppose, fixing a corpus, that we find a non-signalling system. Adding or removing a single word could break the harmony between the distributions, making it signalling. Even more problematic, what would happen if we did it all again, with a much smaller sample? Well, things might not change, but some attested combinations might disappear[15] and in the same breath again change the probability distributions. We thus still need a way of solidifying probabilities in a way that ensures that, even were we to alter the corpus, and change the statistical occurrences of specific combinations, we would still keep a possibly non-signalling behaviour. And this is why the authors considered the Contextuality by Default framework.

6 Contextuality by Default

Explaining in detail the Contextuality by Default framework could be the work of an entire separate paper. We will therefore be content with introducing the general idea behind it. Crucially, note that it "is not a model of empirical phenomena, and it cannot be corroborated or falsified by empirical data. However, it provides a sophisticated conceptual framework in which one can describe empirical data and formulate models that involve random variables."[4]. Its main goal is to "facilitate" non-signallity of systems. To simplify (as Wang et al.[8] never use it in a "table shape"), it can be seen as a hybrid between the possibilistic and probabilistic contextuality models we presented, taking the contextuality criteria for probabilistic contextuality combined with the ease of possibilistic non-signallity. In other words, it is a form of "compression", ensuring the statistical distribution of our probabilistic models do not diverge between interpretations, by homogenising them through the minimum we observe. Then, it is possible to calculate how close a system may be to being contextual. This technique allowed Wang et al.[8] to identify multiple cases of contextuality in natural language, cases obtaining both of our criteria. This is then further expanded upon in [7], where they mark a difference between the amount of contextuality of polysemous and homonymous words, notably through the fact that homonymous verbs seem to disambiguate more that polysemous ones, hinting at a possibly more theoretical interpretation of the quantum method.

7 Critiques and other works

Though rigorous, a few aspects of Wang et al.[8]'s paper can be discussed, and may lead to interesting expansions of the theory. First, it is unclear why the possibilistic approach initially considered did not rely on the same corpus that the probabilistic one did, but instead on gut feeling. The reason one could imagine, is that by giving a pass or fail themselves, the authors ensured elimination of metaphorical interpretations, as they decided to "work with non metaphorical meanings in order to keep the hand annotations of interpretations manageable."[8]. This choice itself, can however be criticised too. In fact, given that the framework here described does not make any distinctions between polysemous and homonymous words, as the

[15] Similarly, if we were to massively enlarge the sample, some unattested combinations might make an appearance.

method could be applied to both, one can wonder why it would not be the same process for metaphors? And in fact, why not expand on similar concepts like homophony and homography, where the ambiguity only arises in one specific way of engaging with the word, namely sound or text? If the discrimination against metaphorical meaning resulted from an ontological claim, a belief about their singularity in theory and incompatibility with the quantum framework, would the same exclusion apply to homophony and homography, as superposition can only be observed under a very specific light? One could also wonder about words gaining or losing interpretations over time. Preexisting words gain slang meanings on daily basis, and lose archaic ones at least as often, but what impact does this have on the contextuality of language? Could it gain or lose contextuality? In a similar way, is it necessary for all interpretations to be accounted for to evaluate the contextuality of a system? If a word is contextual with 3 interpretations, should it not also be so with 2? However, this framework does not ensure this, which seems to indicate that it is still bound to be a method of analysis, and not a claim about the underlying structure of language itself.

Please note that there also exist different applications of quantum theory to linguistics. It has for instance been used to describe vague words (such as "animal", "act", "fruit" or "vegetable"), see [3], and the likelihood for a concept falling under two umbrella terms to end up being expressed via one or the other. Quantum jargon has even been used in less formal manners to describe modalities in language, see [1], in much the same ways that a lot of trending topics get applied for a quick intersectionality flare. Nevertheless, quantum theory appears to be an interesting and vibrant source of inspiration for linguistic studies.

8 Conclusion

In conclusion, the research lead by Wang et al.[8] greatly pushes the idea that quantum theories, and in their case more precisely the notion of quantum contextuality, could have a role to play in linguistic analysis. Although not in its traditional possibilistic or probabilistic forms, but under the adapted guise of the Contextuality by Default framework, it allows for a proper formalisation of lexical ambiguity and the context dependency of language. Other applications of the theory seem as promising, leaving many more research opportunities open, but until they have been thoroughly scrutinised, we will not take the risk to claim the field dead or alive.

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